

HalluXAI: A Claim-Level Multi-Agent Framework for Explainable Hallucination Detection in Large Language Models

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Abstract—Not long ago, verifying information involved searching on Google, cross-checking Wikipedia, or consulting reference books—methods that were slower but provided traceable and reliable sources. Today, users increasingly rely on large language models (LLMs) for instant answers. However, these models often generate confident yet incorrect information (hallucinations), raising serious concerns in critical domains. Existing detection approaches are limited, as they operate at the whole-response level, provide simple yes/no outputs, and depend on closed-source APIs, reducing transparency and trust. We present HalluXAI, an open-source multi-agent framework for claim-level hallucination detection. It decomposes LLM outputs into atomic claims, retrieves supporting or contradicting evidence from external sources, assigns confidence-aware verdicts, and provides explanations with candidate corrections in a unified pipeline. Evaluated on TruthfulQA and HaluEval benchmarks, HalluXAI achieves 87.9% Precision, 83.5% Recall, and F1-score of 85.6%, improving to 97.4% F1-Score in retrieval-augmented settings. Results highlight the critical role of evidence retrieval in ensuring accurate and reliable verification. HalluXAI restores accountability in AI-generated responses by combining speed with transparent, evidence-based validation in a fully open-source and reproducible system.

Index Terms—hallucination detection, explainable AI, multi-agent systems, claim-level verification, retrieval-augmented generation.

I. INTRODUCTION

In today’s time Large Language Models like ChatGPT are used for generating answers, writing summaries, and for conversations in human language [7]. But sometimes these models generate unfactual and fabricated answer which results in hallucination [18]. This is called hallucination. In domains such as medicine, law or science, an incorrect answer is not only an inconvenience but also can be dangerous. There are two main ways hallucinations happen. First, the AI might independently produce incorrect information [14] without any source material — it is a phenomenon known as factual hallucination. Second, even when provided with specific documents for reference (as in Retrieval-Augmented Generation, or RAG systems), the model can still output claims that contradict or lack support from those sources—this is termed RAG hallucination [8]. Most current tools treat both the same way, which is a mistake. The second type can actually be checked against the provided

documents, while the first type requires checking against general world knowledge — these are two different problems that need different solutions. also generated.

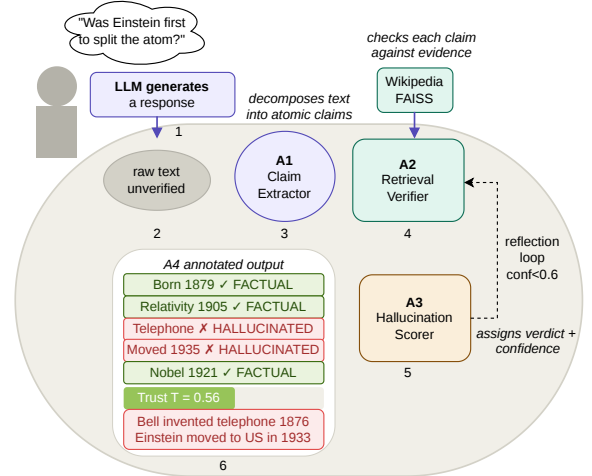


Fig. 1. Architecture of the HalluXAI framework

Current hallucination detection tools also have four major problems:

- 1) They look at the entire AI response as one unit [14] and give a simple yes/no verdict — they don’t tell you which part of the answer is wrong.
- 2) They don’t explain why something is wrong in plain language.
- 3) Many depend on paid, private AI services, making it hard for other researchers to test or reproduce the results.
- 4) They make one pass through the response and never go back to re-check anything they were uncertain about.

Some recent tools like [7] and [2] have tried to add explanations, but breaking responses down into individual claims and explaining each one separately is still largely unsolved. Our system, HalluXAI, is designed to eliminate all this problems. Here is what it does: HalluXAI uses four specialized AI agents

to deliver results i.e., one to break the response into individual claims, one to search for evidence, one to score each claim, and one to write an explanation — giving a detailed and claim-by-claim analysis. This system presented in this paper detects and corrects mistakes which are either internal knowledge based or retrieved information. It shows a better performance when supporting documents are provided and achieves F1-score of 97.4% on the HaluEval dataset. Our system also suggests factual corrections in simple language and about 76.5% of those suggestions are correct. It tries to guide users to make better decisions rather than providing final or definite answer. Fig. 1. shows the architecture of the HalluXAI framework used for detecting hallucination in LLM-generated responses.

The entire system runs on the open-source Mistral 7B model and works on regular consumer hardware, so that researcher can use, test and reproduce on their own.

We have also made the limitations of the system clear. Without supporting documents accuracy drops and makes F1-score 73.8%, whereas when documents are available performance improves and F1-score: reaches to 97.4%. We also added re-checking step for low confidence outputs but it was not much helpful, and we plan to work on it in future. All findings are supported by extensive experimentation and error analysis which we have covered in this paper. Section II of this paper discusses research on hallucination detection, explainability RAG, and multi-agent systems. Section III describes how HalluXAI operates - identifying claims, scoring hallucinations, generating explanations, and an optional re-checking loop. Section IV describes the experimental datasets, baselines and ablations. Section V presents and discusses the results. Section VI highlights current limitations and future directions, and Section VII wraps up with the main takeaways of HalluXAI.

II. RELATED WORK

2.1 Hallucination Detection and Explainability

Early work on hallucination detection used Natural Language Inference (NLI), where a system checks if a statement supports, contradicts, or is unrelated to another statement [15].

An important improvement was introduced in [13], in which responses of AI were divided into two small factual statements and each statement was checked separately. This makes the task of finding incorrect information more accurately little easy. However system uses paid and closed infrastructure, but fails to explain clearly why statements is marked wrong. It also required high computational power and usually gives only yes or no result instead of detailed explanation.

Recent studies have focused on making detection results easier for users to understand. Study [7] improved this area by combining hallucination detection with simple natural-language explanations. Similarly, studies [2] and [10] showed that systems with multiples structured steps perform better than basic methods in terms of reliability and user-friendliness.

HalluXAI further improves this area by introducing an open-source framework that brings together detailed claim checking, evidence-supported validation, and clear explanations within one complete pipeline.

2.2 Verifying AI Outputs Against Retrieved Documents

[8] introduced Retrieval-Augmented Generation (RAG), a method that improves AI responses by providing relevant documents during the generation process. This approach helps reduce factual errors, but hallucinations can still happen when the model misunderstands, changes, or ignores the given information.

Our experimental result also highlight the importance of retrieval. In the HalluXAI ablation study, removal of the retrieval component caused the largest drop in F1- Score (10.5 points) compared to the other changes. This shows that collecting relevant evidence is a key part of the overall system architecture.

2.3 Collaborative Multi-Agent AI Systems

Assigning demanding tasks to a team of tailored AI agents—avoiding overload on a lone model—has validated as a superior tactic. [16] set a seminal standard for agentic interplay, facilitating model-to-model dialogue for joint problem-solving.

Building on this foundation, [18] demonstrated the value of role-specific agents in identifying hallucinations within vision-language models, finding that specialization measurably improved detection outcomes. [15] added another relevant finding — that selectively re-examining outputs the model itself was uncertain about helps contain the spread of errors through a pipeline.

HalluXAI adopts this multi-agent philosophy and applies it to a domain where it has not been fully explored before: explainable, claim-level hallucination detection in text. Each agent in our system handles a specific task, which makes the overall process modular, transparent, and easy to evaluate separately.

III. PROPOSED METHODOLOGY

HalluXAI works through a step-by-step process with a feedback loop created using LangGraph, as shown in Figure 1. It is mainly built to check information in Retrieval-Augmented Generation (RAG) based systems using provided context, even when no reference context is available it can verify factual claims.

3.1 Claim Extractor

The Claim Extractor takes the model’s response and any available reference information and breaks it into small claims where there is only one fact in one sentence. This allows the system to decide which claim is incorrect, rather than saying the whole answer is hallucinated.

3.2 Retrieval Verifier

To verify each claim, the system gathers related information from Wikipedia and available local FAISS database using embeddings. It then compares the evidence with the claim and decides whether the evidence supports it, contradicts with it, or is unrelated. If user gives their own document, the system will prioritize it first instead of searching outside sources.

Because of this, the system works much better when reference documents are available. It achieves F1 score of 97.4% while performance drops to 73.8% when no reference is provided.

3.3 Hallucination Scorer

The Hallucination Scorer checks each claim and classifies it into one of three categories: Factual, Hallucinated, or Unverifiable. It also gives a confidence score between 0 to 1. To make decision, the system follows a fixed order. It gives the highest importance to contradictions found in user-provided documents, with a confidence score of < 0.95 , and then looks at the evidence collected from other sources. Claims with confidence < 0.6 are eligible for a reflection loop. However, empirical evaluation shows negligible impact from this mechanism ($\Delta F1 = 0.0\%$), indicating that the base retrieval and scoring pipeline is already effective. Additionally, the reflection step introduces latency without providing consistent performance gains at the 7B parameter scale.

3.4 Explanation Generator

For each verdict, generates a structured natural language explanation: what is incorrect, a candidate corrected fact, and the supporting evidence source. We frame these as candidate corrections (76.5% LLM-verified accuracy, 26/34 verified) rather than guaranteed facts, acknowledging that corrections may themselves contain errors. For factual claims, the explanation briefly confirms supporting evidence. An executive summary with overall trust score is also generated.

3.5 Reflection Loop (Preliminary)

A conditional edge in the LangGraph pipeline re-routes low-confidence claims to Agent 2 for additional evidence retrieval (up to $R=2$ iterations). Our ablation study shows this mechanism contributed 0.0% F1 improvement on the full 200-sample evaluation, suggesting the loop does not consistently improve results with Mistral 7B at 2048-token context. We include it as an architectural component for future investigation with larger models, but do not claim it as a primary performance contribution.

IV. EXPERIMENT

4.1 Dataset

We tested HalluXAI using two well-known benchmarks. The first is TruthfulQA, which contains 817 questions spread across 38 different categories, all designed around common misconceptions people tend to believe. In this setting, no reference documents are provided — the AI has to rely purely on what it has learned. We picked 200 responses from this dataset, including both correct and incorrect answers. The second dataset used was HaluEval, where each question is paired with a reference documents, making it a context-based fact-checking task. Again, 200 question pairs were selected, equally divided between accurate and hallucinated responses. Across both benchmarks, the system was evaluated on a total of 400 samples. Figure 2. shows the workflow of the HalluXAI system, including claim extraction, verification, scoring and explanation generation for detecting hallucinations.

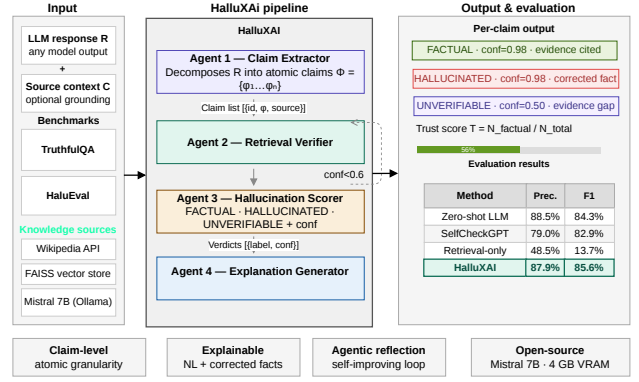


Fig. 2. HalluXAI Workflow

4.2 Baseline Methods

To measure the effectiveness of HalluXAI, we compared it with three simpler baseline methods:

B1 – Zero-shot LLM: The model is given a single prompt to decide whether the response contains hallucination or not, without using any extra steps or external resource.

B2 – SelfCheckGPT-style: For the same query, three responses are generated with randomness in settings. The final decision is based on the result that appears most often among the three outputs.

B3 – Retrieval-only: This methods compares the claim with related wikipedia passages using basic keyword similarity scores, without applying any machine learning reasoning.

All three baselines use the same base model as HalluXAI (Mistral 7B), ensuring a fair comparison.

4.3 Ablation Study Design

To identify the most important components of HalluXAI, we tested three modified version of the system:

V1: The reflection loop was removed, so decisions were made in a single step without re-evaluating uncertain cases.

V2: The retrieval component was removed, meaning the system relied only on the model's internal knowledge and did not use any external evidence.

V3: Claim decomposition was removed, so the complete responses was checked as one unit instead of analyzing each claim separately.

Each modified version was evaluated on 200 samples under the same conditions HalluXAI system.

4.4 Evaluating Explanation Quality

We evaluated the quality of the system's explanations using three methods:

ROUGE-1 F1: Measures how many important words in the explanation match the original claim.

Coverage score: checks whether the explanation clearly discusses the error, provide correct information, and uses proper reasoning language.

TABLE I
ABLATION STUDY – PERFORMANCE RESULTS FOR HALLUCAI VARIANTS

Method	Samples	Accuracy	Precision	Recall	F1 Score	Explainable
B1 Zero-shot LLM	400	85.0%	88.5%	80.5%	84.3%	No
B2 SelfCheckGPT	400	82.0%	79.1%	87.0%	82.9%	No
B3 Retrieval-only	400	49.8%	48.5%	8.0%	13.7%	No
Proposed system	400	86.0%	87.9%	83.5%	85.6%	Yes

Correction Accuracy: Another AI model is used to verify whether the suggested corrections are actually factually correct. This result is reported both as a percentage and as a raw fraction for better transparency.

4.5 How We Measured Performance

We framed hallucination detection as a binary classification problem ("hallucinated" = positive class), tracking Precision, Recall, F1-score, and Accuracy. "Unverifiable" outputs were conservatively scored as no n-hallucinations. Scores are reported per dataset to reflect their distinct setups.

V. RESULTS AND DISCUSSION

A. Baseline Comparisons HalluXAI outperformed three baselines—Zero-shot LLM (B1), SelfCheckGPT (B2), and Retrieval-only (B3)—tested on 400 samples each, all using identical Mistral 7B to guarantee fairness. As shown in Table I, HalluXAI achieved the highest overall performance with 87.9% Precision, 83.5% Recall, and F1-score of 85.6%, alongside an accuracy of 86.0%. The Zero-shot LLM baseline (B1) came closest in performance, achieving 88.5% Precision and 84.3% F1-Score, but relied on a single prompt with no structured pipeline, claim decomposition, or evidence retrieval — and crucially, produced no explanations or corrections. SelfCheckGPT (B2) achieved reasonable Recall of 87.0% through sampling consistency across multiple generations but fell behind in Precision (79.1%) and F1-Score (82.9%), and similarly offered no interpretable output. The Retrieval-only baseline (B3) performed poorly overall, collapsing to an F1-Score of just 13.7% due to near-zero Recall of 8.0%, confirming that lexical overlap against Wikipedia passages alone is wholly insufficient for reliable hallucination detection. HalluXAI was the only system among the four methods that provided claim-level decisions, confidence scores, clear natural-language explanations, and suggested factual corrections. This makes it unique, as it combines strong detection performance with outputs that are actually useful and practical for users.

B. Full Evaluation on 400 Samples

Across all 400 samples, HalluXAI performed better than the baseline methods in both evaluation settings. On HaluEval, where supporting reference documents were available, HalluXAI achieved a Precision of 100%, Recall of 95.0%, and F1-score of 97.4%. On TruthfulQA, where no source documents were given and the system depended only on Wikipedia retrieval, the performance decreased to F1-score of 73.8%. The difference of 23.6 percentage points between the two datasets highlights the challenge of the tasks. When reference context is available, the system can verify claims more accurately using

TABLE II
EVALUATION RESULTS

Dataset	Samples	Accuracy	Precision	Recall	F1 Score	Errors
TruthfulQA	200	74.5%	75.8%	72.0%	73.8%	0
HaluEval	200	97.5%	100.0%	95.0%	97.4%	0
Overall	400	86.0%	87.9%	83.5%	85.6%	0

the provided evidence. In contrast, open-domain fact-checking based on general knowledge is more difficult, especially for a 7B parameter model. The overall figures across both datasets were 87.9% Precision, 83.5% Recall, and 85.6% F1-score. HalluXAI is therefore best suited for RAG-style deployments and should be applied cautiously in purely knowledge-based settings.

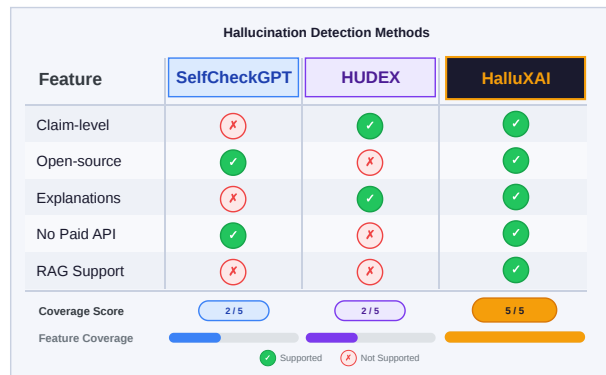


Fig. 3. Feature Comparison

Fig. 3. presents a comparative overview of the key features offered by different hallucination detection methods.

C. Ablation Study Three variants of HalluXAI were tested on 200 samples to isolate the contribution of each core component. Results are summarized in Table IV. Removing evidence retrieval (V2) caused the largest performance drop, with F1-Score falling from 81.9% to 71.3% — a loss of 10.5 percentage points. This is the single largest degradation observed across all ablations and confirms that external evidence grounding is the most critical component of the pipeline. Without it, the system reverts to relying entirely on the LLM’s parametric knowledge, which is demonstrably insufficient for reliable hallucination detection. Removing the reflection loop (V1) produced zero measurable change in F1-Score. We attribute this to the 2048-token context window of Mistral 7B, which is too constrained to benefit from iterative re-examination of low-confidence claims. The mechanism is retained as an architectural component for future investigation with larger models but is not claimed as a performance contributor in the current system. Removing claim decomposition (V3) produced a marginal F1-Score gain of 0.5% but caused Precision to drop from 84.9% to 80.8%. Evaluating whole responses rather than atomic claims recovers a small number of additional hallucinations but introduces more false positives. More importantly, response-level evaluation eliminates the

interpretability advantage that is central to HalluXAI’s design — making claim decomposition essential even where its raw metric impact is modest.

TABLE III
EXPLANATION EVALUATION – PER DATASET

Dataset	Samples	ROUGE-1	Coverage	Corr. Acc.	Avg Words	Errors
HaluEval	100	0.316	0.356	75.0% (15/20)	12.8	0
TruthfulQA	100	0.364	0.505	78.6% (11/14)	17.6	0
Overall	200	0.340	0.430	76.5% 26/34 =	15.2	0

D. Explanation Quality

Explanation Quality Evaluation Explanation quality was tested on 200 samples using three metrics.

Table III shows ROUGE-1 score of 0.340 means there is some word overlap between the explanations and the original claims. This result is expected because correct and incorrect answers often use different words. The coverage score of 0.430 shows that most explanations included three important points: What was wrong, how it can be corrected, and what evidence supports the correction.

The correction accuracy was 76.5%, which means 26 out of 34 checked correction were right. These corrections can be treated as final answers. Human checking it still important.

On average, each explanation had 15.2 words, so the system kept them short and easy to understand. Explanations for TruthfulQA were a little longer (17.6 words) than HalluEval (12.8 words). This happened because general knowledge questions need more explanation, while answers with reference documents are easier to explain.

E. Error Analysis

Three common types of errors were found, and all of them were caused by limitations in the retrieval process.

First, Missed Hallucinations (28 false negatives. Truthful QA): In some cases, Wikipedia returned loosely related but unclear results. Because nothing directly contradicted the claim, the system incorrectly accepted it as true. Second, False Positives (23 cases, TruthfulQA-heavy): Technical terms triggered misleading “contradictions” from semantically close but unrelated snippets. Third, Inaccurate Corrections(8/34 = 23.5%): Overzealous fixes that mangled partially correct claims or swapped one inaccuracy for another.

F. Overall Assessment

HalluXAI’s primary contribution is not only for any single metric but it is the combination of capabilities which delivers within one open-source system: atomic claim-level verification, confidence-scored verdicts, natural language explanations, candidate corrections, and full reproducibility on consumer hardware. It achieves state-level performance in document grounded settings (F1-Score: 97.4%) and competitive performance in open-domain settings (F1-Score: 73.8%) for a 7B parameter model. Compared to HUXE and similar tools, HalluXAI is the only system which make all of these

TABLE IV
ABLATION STUDY - PERFORMANCE RESULTS FOR HALLUXAI VARIANTS

Variant	Samples	Accuracy	Precision	Recall	F1-Score	Delta F1-Score
Full HalluXAI	200	82.5%	84.9%	79.0%	81.9%	–
V1 - No reflection loop	200	82.5%	84.9%	79.0%	81.9%	+0.0%
V2 - No retrieval	200	75.5%	85.9%	61.0%	71.3%	-10.5%
V3 - No claim decomp.	200	82.0%	80.8%	84.0%	82.4%	+0.5%

capabilities freely available without dependence on any proprietary APIs — directly lowering the barrier for reproducible hallucination research. Fig. 4 is shown pictorial representation of performance of HalluXAI over other baseline methods.

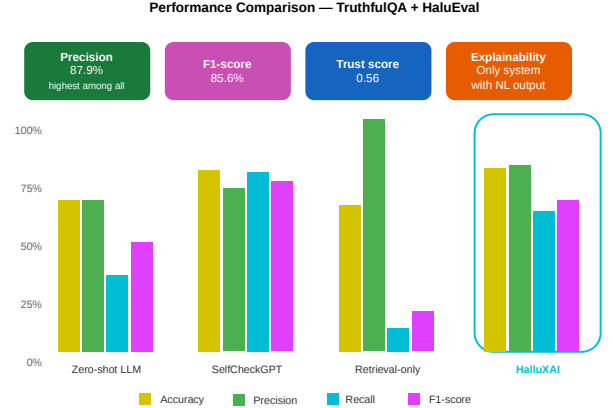


Fig. 4. Performance Comparison

VI. LIMITATION

There are four honest limitations to acknowledge:

First, it needs source documents to work best: Accuracy drops without reference materials. So it should be used with caution in standalone face-checking situations.

Second, it struggles with conceptual errors: if Wikipedia lacks direct evidence of particular claim. It is also hard to detect when the response is complicated or about unusual situations.

Third, corrections are not always right, about 1 in 4 fixes are wrong, so they should be used as a starting point for human review, not taken as authoritative.

Fourth, Reflection loop is not contributing much. This feature which is designed to re-examine uncertain predictions showed no benefit at the current model size. A larger model with more memory may change this.

VII. CONCLUSION

HalluXAI is a multi-agent framework that breaks LLM responses into atomic claims, identifies hallucinations with confidence scores, delivers plain-language justifications, and suggest corrections. Evaluated on 400 samples, it reaches F1-score of 85.6%, rising to 97.4% in RAG scenarios(HaluEval). The ablation study makes clear evidence that evidence retrieval is the most important component. Explanation quality checks show 76.5% correction accuracy. Running entirely on the free,

open-source Mistral 7B model on regular hardware. Now any researcher can make this kind of analysis using HalluXAI without needing any paid API access. The main focus of the system is to make hallucination detection transparent, explainable and easy to verify. In the future we plan to improve system with better retrieval methods, efficient verification components, test reflection loop on larger models and expand the system to other languages and specialized areas.

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